



Sportify

Cloud-Native Multi-Sport Club Management System (MSCMS)

A modern multi-sport club should not run on a dozen apps and a hundred spreadsheets. Sportify brings squads, live matches, medical, plan-driven training, real competitions, analytics and fans into one secure, role-aware platform.

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The Problem

Why a multi-sport club struggles without a unified platform

■ Scattered Tooling

Splinters club operations across spreadsheets, WhatsApp threads and paper forms, with no shared system.

■ Blind Injury Tracking

Records injuries in isolation, with no link to player availability or selection decisions.

■ Role-Blind Access

Exposes sensitive data through shared logins, with no per-role permissions or audit trail.

■ Unplanned Training

Schedules sessions ad hoc, with no attendance, drill rating or training-load tracking.

■ Manual Match Ops

Forces staff to log line-ups, events and scores by hand, inviting delays and human error.

■ No Single Truth

Leaves standings, squads and fixtures duplicated and contradictory across teams and sports.

■ Zero Analytics

Offers no performance insights or match-outcome predictions to guide tactical choices.

■ Slow Deal Flow

Drags scouting reports and sponsor negotiations through disconnected emails and back-and-forth.

Project Overview and Purpose

What Sportify is, how it works, and who it helps



What it is

A cloud-native platform that unifies every operation of a multi-sport club into one secure, role-aware system, replacing scattered spreadsheets, paper files and chat threads.



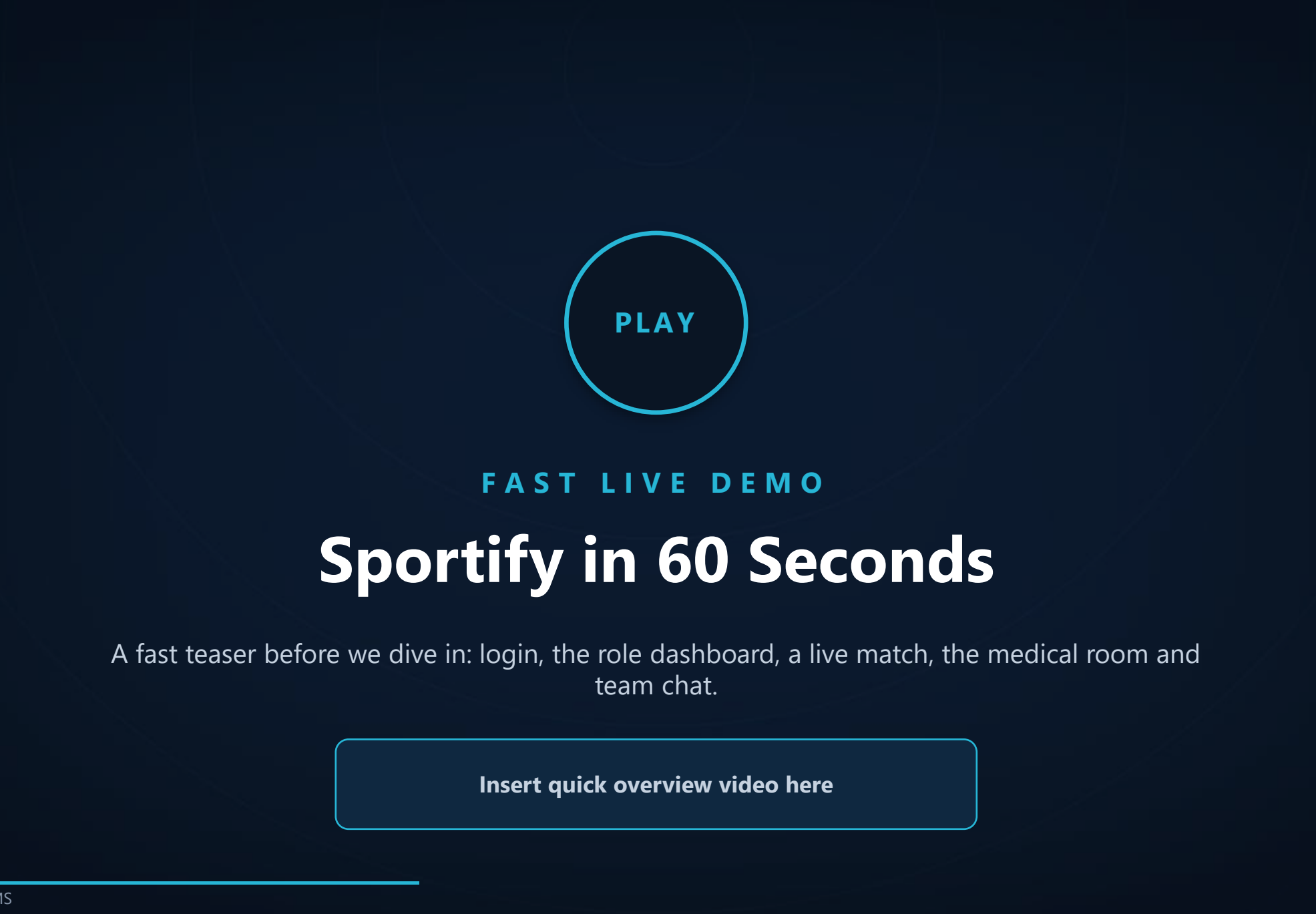
How it works

A Next.js dashboard talks to one JWT-secured API gateway, which routes to six Spring microservices and a Python ML and AI layer, all kept in sync by an event backbone.



How it helps

Every role gets exactly the screen it needs, matches run live, injuries gate selection automatically, real competition data stays the single source of truth, and analytics drive decisions.



PLAY

FAST LIVE DEMO

Sportify in 60 Seconds

A fast teaser before we dive in: login, the role dashboard, a live match, the medical room and team chat.

Insert quick overview video here

The Solution

How Sportify answers each problem: old playbook to cloud-native one

■ Spreadsheets → Cloud Platform

Replaces scattered, siloed spreadsheets with one unified cloud-native platform that centralizes every workflow.

■ Paper Files → Digital Journey

Turns loose paper medical records into a staged digital injury lifecycle tied to each player's profile.

■ No Live Ops → Match Engine

Upgrades silent match days into a broadcast-style live engine streaming events, line-ups and a running clock.

■ Manual Standings → Real Data

Swaps hand-typed tables for live standings and fixtures ingested automatically from the football-data.org API.

■ Open Access → RBAC

Ends 'everyone sees everything' by enforcing Keycloak role-based access across players, staff and managers.

■ Gut Feel → Analytics & ML

Replaces instinct with data dashboards and a Python ML service that predicts outcomes and player insights.

Project Goals

What Sportify set out to achieve

- **Unify Club Operations:** Consolidates matches, medical, training, scouting, messaging and finance into one role-aware platform.
- **Run Matches Live:** Streams each fixture in real time with an accurate, minute-by-minute event timeline and live clock.
- **Drive Injury Recovery:** Guides every injury from diagnosis through rehab to a final passing fitness test before return-to-play.
- **Anchor a Single Source of Truth:** Makes the Spring Boot back-end the authoritative store for real fixtures, results and league standings.
- **Enforce Secure Access:** Applies Keycloak-backed role-based access control across every service, route and action.
- **Surface Predictive Insights:** Delivers performance analytics and ML-driven predictions to inform line-up, scouting and tactics.

System Architecture

A cloud-native Spring Cloud microservice mesh behind one JWT-secured gateway

■ Next.js Front-End

Serves the React dashboard and talks to the platform through a single API Gateway origin.

■ Spring Cloud Gateway

Verifies Keycloak JWTs and load-balances every request to the right service on port 8080.

■ Six Domain Services

Splits into user (8081), player (8084), training-match (8087), medical (8086), reports (8083) and notification (8085), each with a private database.

■ Eureka + Config

Registers services in Eureka (8761) and feeds them centralized settings from the Config Server (8082).

■ Keycloak Identity

Issues and signs OAuth2 tokens for the mscms realm that every service validates as a resource server.

■ Kafka Event Backbone

Streams domain events asynchronously via an outbox publisher so notifications and analytics never block.

■ FastAPI ML Engine

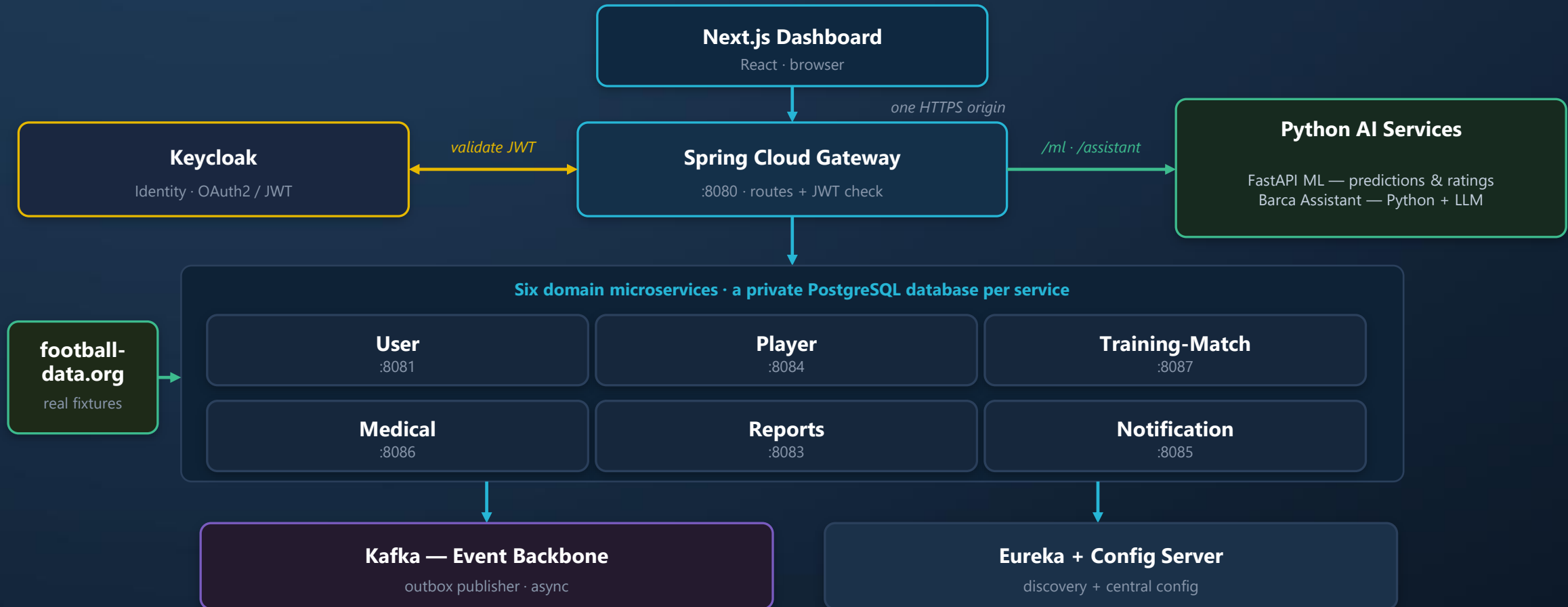
Exposes Python match-prediction and player-rating models that the gateway routes to under /ml.

■ football-data.org Feed

Pulls real La Liga and Champions League fixtures and standings on a scheduled sync into the match service.

Architecture at a Glance

Clean layers · one gateway · Keycloak as a side-guard · Python AI as separate services



Enterprise-Grade Tech Stack

Proven, production-grade technology at every layer



Back-End

Java 21 / Spring Boot 3.5
Spring Cloud Gateway + Eureka
PostgreSQL 14 (DB per service)
Apache Kafka (eventing)



Front-End

Next.js 16 (App Router)
React 19 and Tailwind v4
JWT-secured API client



Infrastructure and AI

Keycloak 26 (Identity / JWT)
Docker (containerized stack)
Python FastAPI (ML models)
Python + LLM (Club Assistant)

User Roles and Access

Eleven role surfaces: each one's value, then its own screen.

Administrator

Role value and core responsibilities



What this role drives — manages all accounts and roles and oversees the platform with unrestricted access.



Key access: every dashboard route, including admin-only User Management.

Administrator — what they see

Insert the Administrator dashboard screenshot here

Full access: every section, plus admin-only User Management.

Sport Manager

Role value and core responsibilities



What this role drives — runs a sport end-to-end, from squad and staff to transfers and sponsor deals.



Key access: Club Hub, Teams, Staff, Players, Matches, Scouting, Contracts and Sponsors.

Sport Manager — what they see

Insert the Sport Manager dashboard screenshot here

Squad, staff, matches, scouting, contracts and sponsor deals.

Team Manager

Role value and core responsibilities



What this role drives — coordinates a team's roster, training schedule, fixtures and contract logistics.



Key access: Club Hub, Teams, Staff, Players, Training, Matches, Contracts, Reports.

Team Manager — what they see

Insert the Team Manager dashboard screenshot here

Roster, training schedule, fixtures and contracts.

Head Coach

Role value and core responsibilities



What this role drives — leads coaching, sets up training and line-ups, and reviews performance and fitness.



Key access: Players, Teams, Training, Matches, Medical, Analytics, Reports, Alerts.

Head Coach — what they see

Insert the Head Coach dashboard screenshot here

Training, line-ups, matches, medical and analytics.

Doctor

Role value and core responsibilities

- **What this role drives** — diagnoses each injury and prescribes its treatment, opening the player's medical case and driving the early stages of the injury cycle.
- **Key access:** Medical (Injuries, Diagnoses, Treatments), Players, Training and Alerts.

Doctor — what they see

Insert the Doctor dashboard screenshot here

Injuries, diagnoses and treatments for a player's medical case.

Physiotherapist

Role value and core responsibilities

- **What this role drives** — builds and manages the rehabilitation plans and tracks each player's recovery progress back toward full fitness.
- **Key access:** Medical (Rehabilitation, Recovery), Players and Training.

Physiotherapist — what they see

Insert the Physiotherapist dashboard screenshot here

Rehabilitation plans and recovery tracking.

Fitness Coach

Role value and core responsibilities

- **What this role drives** — creates, assigns and tracks fitness tests, clearing a recovering player for return-to-play only once a final fitness test is passed.
- **Key access:** Training, Fitness Tests, Players and Analytics.

Fitness Coach — what they see

Insert the Fitness Coach dashboard screenshot here

Fitness tests: create, assign, track and clear return-to-play.

Performance Analyst

Role value and core responsibilities



What this role drives — analyzes match and training data and runs AI predictions and ratings.



Key access: Analytics, Training Analytics, Reports, Matches, Players and the ML Predictor & Rating tools.

Performance Analyst — what they see

Insert the Performance Analyst dashboard screenshot here

Analytics, reports and the ML predictor and rating tools.

Scout

Role value and core responsibilities

- **What this role drives** — tracks and evaluates prospects, logging scouting reports against players and fixtures.
- **Key access:** Scouting & Scouting Ops, Players, Matches, Competitions, Head-to-Head, Reports.

Scout — what they see

Insert the Scout dashboard screenshot here

Scouting, prospects, matches and head-to-head.

Player

Role value and core responsibilities

- **What this role drives** — follows the club and a personal schedule, viewing fixtures, results, assigned training sessions and personal medical and fitness status in a focused, view-only surface.
- **Key access:** read-only Club Hub, Players, Matches, Competitions and Head-to-Head, plus the player's own Training and Medical.

Player — what they see

Insert the Player dashboard screenshot here

A read-only club, matches, training and own-medical view.

Fan

Role value and core responsibilities

- **What this role drives** — the public face of the platform. Browses club news, squads, fixtures and live results; follows competitions and the trophy room; engages with the fan community; shops the official store; and bids against other fans in live auctions for exclusive, signed memorabilia.
- **Key access:** Club Hub, Trophy Room, Competitions, Matches and Head-to-Head, plus the Club Store, the Auctions house and the News Feed.

Fan — what they see

Insert the Fan dashboard screenshot here

Club hub, competitions, store, auctions and the news feed.

Key Features

Each capability explained, then shown live.

Role-Aware Dashboards

One adaptive surface per role, from administrator to fan.

- **Adaptive surface** — Every menu, page and action is tailored to the signed-in role.
- **Enforced twice** — A shared permissions map gates the sidebar and the server-side route guard.
- **Nine role surfaces** — From administrator to fan, each account sees only what it should.

Role-Aware Dashboards · In Action



Embed the Role-Aware Dashboards demo video here

The admin or coach dashboard with its sidebar.

Multi-Sport and Multi-Team

Sports, teams, rosters, contracts and transfers in one place.

- **Many sports** — Football, basketball, handball and more, all in one system.
- **First and reserve squads** — Each sport runs a senior team plus a reserve or academy team.
- **Full squad data** — Players, staff, contracts and transfers managed per team.

Multi-Sport and Multi-Team · In Action



Embed the Multi-Sport and Multi-Team demo video here

The Teams and Sports page with multi-sport squads.

Real Competitions

Live La Liga and the new-format UEFA Champions League.

- **La Liga Live** — Shows live La Liga standings and fixtures, with results that update automatically as matches are played.
- **New-Format Champions League** — Renders the redesigned UCL as one 36-team league-phase table with qualification zones, then unlocks the knockout bracket.
- **World Cup 2026** — Presents the 48-team World Cup across 12 group tables and a knockout bracket, seeding the qualified teams into the Round of 32 when the draw is TBD.
- **Head-to-Head** — Builds FC Barcelona's all-time win-draw-loss record, goals and win-rate against every opponent from finished matches.

Real Competitions · In Action



Embed the Real Competitions demo video here

The La Liga table or the Champions League league-phase.

Live-Match Engine

A broadcast-style match-day cockpit, run minute by minute.

- **Phase State Machine** — Walks every match through scheduled → first half → half time → second half → full time, auto-going-live at kickoff.
- **Phase-Bound Events** — Records goals, cards, subs and injuries against a live clock, clamping each event's minute to the phase in play.
- **Knockout Decider** — Triggers two extra-time periods and an alternating penalty shoot-out only when a true knockout tie is still level.
- **Live Persistence** — Streams every score change, event and status flip to the training-match service so the timeline survives a reload.

Live-Match Engine · In Action



Embed the Live-Match Engine demo video here

A live match: clock, events and the add-event panel.

Advanced Injury Management

A medically-governed return-to-play pipeline.

- **Staged Injury Lifecycle** — Drives every injury through REPORTED → DIAGNOSED → TREATING → RECOVERING → RECOVERED as the status that gates each next action.
- **Single Source of Truth** — Makes the medical service the one authority on availability, so card, stepper and squad all read the same status.
- **Auto-Removal When Injured** — Pulls an injured player out of match line-ups, training and attendance the moment the injury is logged, and notifies the team.
- **Mandatory Fitness Test** — Requires a passing return-to-play fitness test before the player flips to RECOVERED and is restored as AVAILABLE.

Advanced Injury Management · In Action



Embed the Advanced Injury Management demo video here

A player's staged medical recovery journey.

Plan-Driven Training

From season plan to scored session, one squad at a time.

- **Training Plans** — Anchors one plan to a single team and sport, with a start-to-end date range, training type, goals and focus that every session inherits.
- **Scheduled Sessions** — Places each session at its own date and time inside the plan's range, then lists them in chronological order with a done count.
- **Ordered Drills** — Stamps each drill with a category, duration and sequence, and rejects the session if the drills' total minutes reach the session length.
- **Squad Attendance** — Records a per-player status and offers only non-injured players who belong to that plan's team for call-up.

Plan-Driven Training · In Action



Embed the Plan-Driven Training demo video here

A training plan with its sessions, drills and attendance.

Scouting and Sponsorship

Scout reports on prospects and sponsor-offer negotiation.

- **Sign-or-pass reports** — Scouts log structured reports (country, position, club, sport), grouped by sport.
- **Sponsor negotiation** — Offers move through pending, counter-offer and accepted states.
- **Auto-routed** — A new report emails and alerts the sport manager automatically.

Scouting and Sponsorship · In Action



Embed the Scouting and Sponsorship demo video here

A scouting report and a sponsor-offer negotiation.

Team Messaging

Club-wide group chat, delivered to every member.

- **Group delivery** — A message posted to a group reaches every member of that group.
- **Reactions and replies** — Threaded replies and reactions keep conversations organized.
- **Team-scoped** — Staff and players only; fans stay on the spectator surface.

Team Messaging · In Action



Embed the Team Messaging demo video here

A team-chat group conversation.

Analytics and ML Predictions

Match-outcome and player-rating predictions on real data.

- **KPI dashboards** — Aggregates team, player and training KPIs from real records.
- **Match predictor** — A Python and FastAPI service forecasts match outcomes.
- **Player ratings** — An ML 0-100 rating informs selection and scouting.

Analytics and ML Predictions · In Action



Embed the Analytics and ML Predictions demo video here

The Analytics dashboard or the ML predictor and rating.

Live Notifications

The whole club, kept in the loop in real time.

- **Top-bar bell** — Injuries, goals, transfers and report-ready events surface instantly.
- **Event-driven** — Kafka events fan out to the notification service, with email.
- **Alerts center** — A full history of everything that happened.

Live Notifications · In Action



Embed the Live Notifications demo video here

The notifications bell open, or the Alerts center.

AI Club Assistant

A Phase-1 intelligent assistant, the Barca Assistant.

- **Domain-specific Q&A** — Answers questions about the club: sports, players, trophies and fixtures.
- **Bilingual replies** — Responds naturally in more than one language for fans and staff.
- **Python micro-service + LLM** — A dedicated FastAPI service backed by an LLM. Phase 1, built to grow.

AI Club Assistant · In Action



Embed the AI Club Assistant demo video here

The Barca Assistant answering a club question.

Future Work

Two phases: a fan store and auctions, then a full engagement economy.

Phase 1 — Club Store and Auctions

Turning fans into customers

■ Official Club Store

An online merchandise store for fans: kits, training gear and memorabilia, with secure checkout and order tracking.

■ Bidding / Auction System

Timed auctions for exclusive, one-of-a-kind items where fans bid against each other in real time, with a live highest-bid, anti-snipe extension and a winner checkout.

■ Exclusive Signed Items

The auction's headline lots: gear signed by players, the coach or the club president. Scarcity drives the highest bids and margins.

■ Built on the Platform

Reuses the existing identity, notifications and payment rails, so the store and auctions plug straight into the current system.

Phase 2 — Fan Engagement and Monetization

Scaling reach and revenue

■ Fans and Sponsors Ad Board

A tailored advertisement board with placed, targeted slots that fans and sponsors can buy, turning audience reach into recurring revenue.

■ Club News Feed and Page

A comprehensive news feed and club page for official updates, match recaps, signings and announcements: the fan home of the platform.

■ Native Mobile Apps

iOS and Android apps with push notifications so fans, staff and players stay engaged on the move.

■ Deeper ML and Wearables

Richer predictions fed by live in-match features and wearable biometrics streamed into player profiles.

What I Got from This Project — Financial Section

Five distinct, realistic revenue streams — each with its own payer and mechanism

■ B2B SaaS Subscription

The core, recurring revenue: clubs and academies pay a tiered monthly or annual subscription to run their operations on the platform. Payer: clubs.

■ Auction Commission

A percentage cut of every completed sale in the fan bidding system — a purely transactional fee on exclusive, signed memorabilia. Payer: bidding fans.

■ Store Sales Margin

A commission on fixed-price official merchandise sold through the club store — distinct from auctions in mechanism and catalog. Payer: shopping fans.

■ Sponsorship & Targeted Ads

Paid, targeted placements and sponsor packages on the fan-facing surface (ad board and news feed), priced by audience reach. Payer: sponsors and advertisers.

■ Premium Fan Membership

An optional recurring fan tier: early auction access and exclusive content — a different payer and cadence from the club SaaS. Payer: fans.

Sustainability and Business Model

Recurring base, transactional upside, one platform for many clubs

- **Recurring base** — SaaS licensing to clubs and academies gives predictable, recurring revenue.
- **Transactional upside** — Store sales, auction commission and ad placements scale with fan engagement.
- **Low marginal cost** — One cloud platform serves many clubs, so each new customer is mostly profit.
- **Reinvestment loop** — Revenue funds richer ML, mobile apps and more sports, which attract more clubs and fans.
- **Project to product** — A graduation project with a credible, staged path to a real, sustainable business.

Conclusion

The Big Picture

- Turns a club's scattered, manual operations into one secure, real-time platform.
- Real value for every audience: staff run the club, players stay informed, fans stay engaged.
- Real competitions, a live-match engine and a complete medical workflow on real data.
- An extensible foundation ready for a fan store, auctions and a wider sporting world.

The Technology and Why

Spring microservices — independent scaling and a database per service: clean separation of concerns.

Gateway + Keycloak — one secure entry point with role-based access enforced on every request.

Apache Kafka — event-driven, so notifications and analytics never block the user.

Next.js / React — a fast, role-aware UI delivered from a single API origin.

Python FastAPI + LLM — ML predictions and a club assistant, kept as separate services.

Docker — the whole stack reproducibly containerized for deployment.

With sincere gratitude to our supervisor Dr. Shaimaa Saber, the Faculty of Computers and Information, and Menoufia University.

Team Members

Supervised by Dr. Shaimaa Saber



Mohamed Salem



Mohamed Abdellatief



Mariam Mohamed Ahmed



Eman Ahmed Elboghdady



Khaled Mohamed



Shahd Abdelaziz

Menoufia University · Faculty of Computers and Information · Graduation Project 2025-2026

Thank You

Questions and discussion welcome

Sportify · MSCMS — a cloud-native multi-sport club management platform
